Christoffel Symbols: Unified View (Polar Coordinates)

1. Coordinate System

$$(r, \theta)$$

Metric:

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \quad g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix}$$

2. Summary of Nonzero Christoffel Symbols

$$\Gamma^r_{\theta\theta} = -r \quad \Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}$$

3. Comparison of the Three Approaches

Method	**Key Formula**	**Interpretation**
Metric (algebraic)	$\Gamma^k_{ij} = \frac{1}{2} g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$	Derived purely from the metric. Measures how the inner product changes across space.
Basis derivatives (geometric)	$\frac{\partial \mathbf{e}_i}{\partial x^j} = \Gamma^k_{ij} \mathbf{e}_k$	Christoffel symbols describe how the coordinate basis vectors change from point to point.
Covariant derivative / projection	$\Gamma^k_{ij} = \mathbf{e}^k \cdot \frac{\partial \mathbf{e}_i}{\partial x^j}$	Christoffel symbols are components of the derivative of a basis vector projected onto the dual basis.

4. Equivalence of the Methods

All three approaches give identical results because:

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j$$

Taking derivatives:

$$\partial_k g_{ij} = \left(\frac{\partial \mathbf{e}_i}{\partial x^k} \right) \cdot \mathbf{e}_j + \mathbf{e}_i \cdot \left(\frac{\partial \mathbf{e}_j}{\partial x^k} \right)$$

Substituting:

$$\frac{\partial \mathbf{e}_i}{\partial x^k} = \Gamma^m_{ik} \mathbf{e}_m$$

yields the metric formula.

5. Geodesic Equations

A geodesic is defined by:

$$\frac{d^2x^k}{d\lambda^2} + \Gamma^k_{ij}\frac{dx^i}{d\lambda}\frac{dx^j}{d\lambda} = 0$$

6. Explicit Equations in Polar Coordinates

Radial equation $(k = r)$

$$\frac{d^2r}{d\lambda^2} + \Gamma^r_{\theta\theta}\left(\frac{d\theta}{d\lambda}\right)^2 = 0$$

$$\frac{d^2r}{d\lambda^2} - r\left(\frac{d\theta}{d\lambda}\right)^2 = 0$$

Angular equation $(k = \theta)$

$$\frac{d^2\theta}{d\lambda^2} + 2\Gamma^\theta_{r\theta}\frac{dr}{d\lambda}\frac{d\theta}{d\lambda} = 0$$

$$\frac{d^2\theta}{d\lambda^2} + \frac{2}{r}\frac{dr}{d\lambda}\frac{d\theta}{d\lambda} = 0$$

7. Interpretation

- The radial equation:
  $$\frac{d^2r}{d\lambda^2} - r\dot{\theta}^2 = 0$$
  represents the familiar "centrifugal" term.
- The angular equation:
  $$\frac{d^2\theta}{d\lambda^2} + \frac{2}{r}\dot{r}\dot{\theta} = 0$$
  encodes conservation of angular momentum.

8. Final Insight

Geodesics are curves whose tangent vector is parallely transported:

$$\nabla_{\dot{\gamma}}\dot{\gamma} = 0$$

Christoffel symbols act as the correction terms that compensate for the curvature of the coordinate system.